

BeeNow – A Remote Beehive Monitor using IoT

Chitra Venugopal
Department of Electrical Engineering
and Renewable Energy
Oregon Institute of Technology
Oregon, USA
chitra.venugopal@oit.edu

Bhuvaneswari Thangavel
Faculty of Engineering and Technology
Multimedia University
Malaka, Malaysia
t.bhuvaneswari@mmu.edu.my

Nathaniel Leascu
line Department of Electrical
Engineering and Renewable Energy
Oregon Institute of Technology
line 4: Oregon, USA
nathaniel.leascu@oit.edu

Abstract— Beekeeping is a time intensive and laborious task, especially in the realm of commercial beekeeping. In his paper, the design and building of prototype model of a remote beehive monitor that a beekeeper could use to more efficiently manage a hive and garner useful data on the hive's health is discussed. The main goal was to track their weight and temperature over time. Additionally, orientation was important to track in case the hives were pushed over or moved without the beekeeper's knowledge. Finally, to further assist the beekeeper, the remotely feed the hives and automate the process without the beekeeper needing to be on site is discussed. The design methodology of the hardware and system is explained. The results obtained and future directions for more accurate automation of the system is also discussed.

Keywords—Beehive monitoring, Load Cell, Microcontroller MPU 6050, GSM, Google worksheets, temperature monitoring system

I. INTRODUCTION

Beekeeping is a labor and time-intensive venture. Commercial beekeepers typically manage hundreds if not thousands of hives. These hives are moved from location to location locally and even nationally, depending on which plants and crops are in pollination. Typically, each hive is manually opened to inspect honey and brood production. In areas where there is limited pollen, supplemental food is provided to feed and maintain the health of the hive.

The research and development conducted for this paper explored ways to connect today's beekeepers to their hives by integrating the hives with an embedded system and transmitting the data back to the beekeeper. The data collected from these sensors equips the beekeeper to efficiently manage the hives by allowing them to identify which hive needs attention based on weight, orientation, and temperature. The object of this research is to show that technology can not only save a beekeeper time, but it can also help him garner useful data from his hive. This enables a beekeeper to manage and keep each hive healthy and thriving.

The hypothesis of this study is that by using a MPU 6050, a load cell and relay, the beekeeper will be able to use this prototype as a means to design and build a smarter set of beehives that can be constantly monitored from anywhere. The specific objectives are to demonstrate collecting weight, orientation, temperature data and send that back to the user.

There are several researches conducted in monitor beehives. Swarming occurs when the bees' current hive is overcrowded, and they are no longer able to regulate the conditions in that hive. A microphone was also attached to monitor the sound bees were making to garner and analyze their temperament.[3][4][7][10][11]. A work that was strictly focused on the well-being of honeybees added a humidity

sensor along with the temperature sensor to detect potential swarm events due to drops in temperature is discussed in [9].

The performance of temperature difference sensors is essential in beehive monitoring. There are several similar temperature monitoring system was found in the literature. Firemen can use them to quickly access the intensity of a burning fire in a room [25]. With beekeeping this information is relevant because bees maintain a constant temperature in their hive and could serve early warning signs of a problem.

In this research MPU 6050 is used as a temperature and tilt sensor. There are several other methods are used similar to [20] where tilt sensors was discussed to detect the slope in their landslide early warning system. In other research it is used as a fault detection system for UAVs [18]

Embedded system and beehive monitoring are the only systems that send out data to users. Engineers in India have developed a system to better manage waste dumpsters throughout the city by using an embedded system to monitor waste and co2 levels [28]. Similar system is adopted in this research.

Load cells are typically used as scales in [17]. The work is being done to use them for the rehabilitation of stroke patients. The work being done by Lee, Kim, Park, and Bae in Korea provides patients with measurable results in the recovery process. These patients were given a wearable hand exoskeleton that gave them a measurable and quantitative way to measure their recovery progression. In another research [15] the researcher analyzed and demonstrated the use of load cell for posture correction at the home or office. Three load cells were built into a chair and paired with an ECG to determine a user posture based on distribution of weight and heart rate. Similar approach is used in this research to observe the weight of the beehives.

This paper explains the design and analysis of a fully functional beehive monitor. This monitor will be able to text the user meaningful alerts and record all data to the cloud to monitor metrics in real time. The prototype will be able to measure weight, measure temperature and track orientation data and notify the intended user.

II. DESIGN METHODOLOGY

A. Software Design

The beehive monitor tracks hive health by measuring temperature, weight and acceleration. Besides measuring sensor values, a relay can be turned on remotely to run a pump that can be used to feed and top off supplemental food for the hives. The sensors used are a load cell in combination with HX711 to track weight. An MPU6050 measures acceleration and temperature. And a relay, in conjunction with a peristaltic pump, is used to pump fluid.

The remote beehive monitor (RBM) consists of a microcontroller that collects data from a HX-711Board and a MPU6050. Based on the weight value from the load cell, it turns on and off a relay that drives a peristaltic pump used to supply the hive with supplemental food. The MPU6050 is used to measure temperature and direction of acceleration. And the HX-711 is used to amplify the measured signal that occurs when force is placed on the load cell. The block diagram of the system is shown in Fig.1.

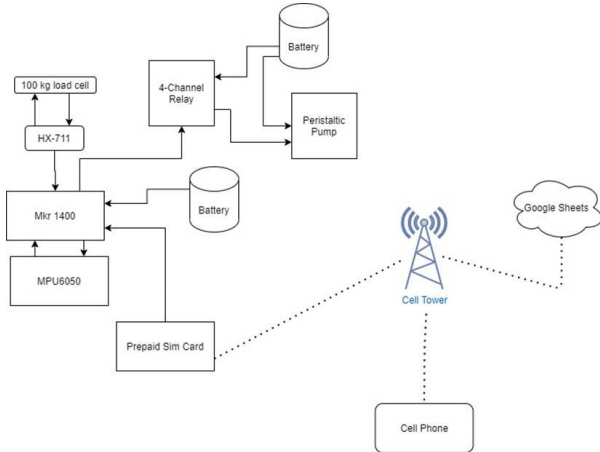


Fig. 1. Block Diagram of the system

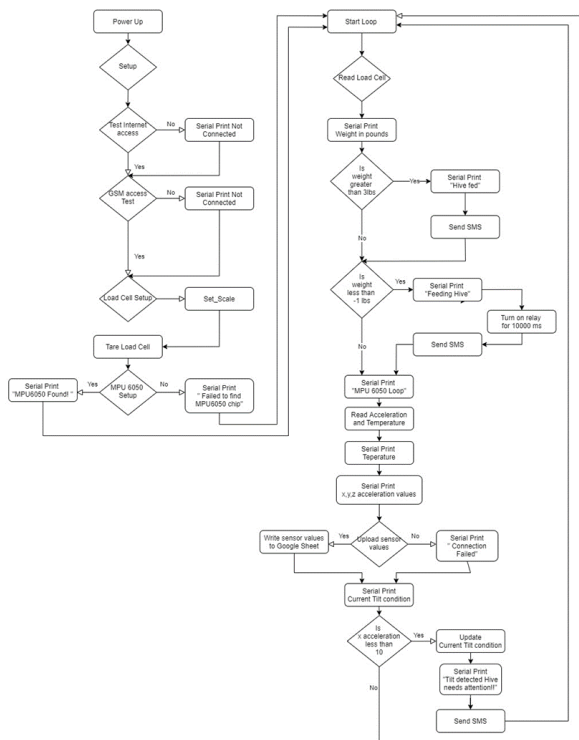


Fig.2 Flow chart of software Design

III. SYSTEM DESIGN

The beehive monitor is a combination of sensors and actuators that are all controlled by a microcontroller. A prototype of a remote beehive monitor is designed with the following essential features:

1. Weight sensor
2. Orientation sensor

3. Temperature sensor
4. Relay and pump
5. Google sheet integration to log and visualize data
6. SMS text notifications over a cellular network

The load cell circuit is used to measure weight. The MPU6050 is used to track orientation and temperature. The relay circuit is used to drive a peristaltic pump used to feed the bees. The MKR 1400 uses a gsm sim card to send text alerts and connect to the internet to upload sensor readings to a Google sheets document. The relay is triggered by a [high or low] signal from the microcontroller and the relay switches the peristaltic pump circuit which is driven by the 9V battery. The MPU 6050 is a 6-axis motion detecting sensor with temperature detect on-chip. It communicates with the microcontroller using an I2C bus. It consists of a 3-axis accelerometer, 3-axis gyroscope and an on-chip temperature sensor. In this research there are only two of those sensors used: the accelerometer to determine current orientation and to detect if the hive has been knocked over, and the temperature sensor intended to track the internal beehive temperature over time.

IV. RESULTS AND ANALYSIS

The MKR GSM 1400 plays a vital role in this design. The flowchart of the software used is shown in fig. 2. It has a built in U-blocks SARA-U2 cellular module. This was used in conjunction with a good2go prepaid sim to add SMS messaging and internet connectivity.

The prototype design is shown in fig. 3. Sensor data was collected, stored and then uploaded to a Google spreadsheet. The uploaded data was visualized using three charts: one for weight, the second for temperature, and the third for orientation which was updated every time the microcontroller uploaded and updated the sheet with new sensor values. This is shown in fig. 4. To test and verify SMS functionality, conditions are set on certain weight and orientation conditions. When these conditions are satisfied, a notification is sent to his phone. The sensor values uploaded to google sheets is shown in fig. 5

The prototype design is shown in fig. 3.

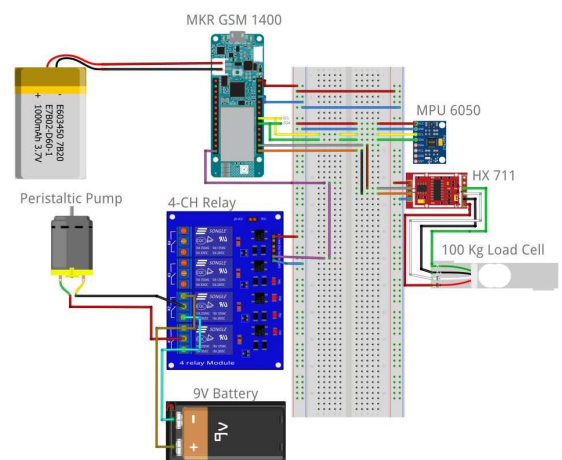


Fig.3 Hardware Design

To test the sensors and load cells circuit, C++ script and library functions are used. The weight data from the load cell

was obtained via the Hx711. The design was tested with practical loads from 2.5lb to 5lb weights.

The output of the sensors was observed in three ways: 1. on the Arduino web editor serial monitor, 2. on the uploaded sensor data and chart on the Google sheet, and 3. via the SMS notification.

	A	B	C	D	E	F	G	H	I	J	K	L
1	weight											
2	0.02	83.44	10.26	0.06	-1.4							
3	0.01	83.69	10.25	0.06	-1.48							
4	0.01	83.69	10.23	0.06	-1.5							
5	0.02	83.95	10.26	0.11	-1.48							
6	0.02	83.78	10.24	0.04	-1.51							
7	0.02	83.86	10.27	0.02	-1.49							
8	0.02	83.96	10.19	0.08	-1.52							
9	0.02	83.69	10.26	0.02	-1.48							
10	0.02	83.89	10.24	0.07	-1.58							
11	0.03	83.65	10.27	0.01	-1.56							
12	0.03	83.86	10.25	0.1	-1.56							
13	0	83.95	2.53	0.15	-10.45							
14	0	83.69	3.37	-8.53	-9.93							
15	0.01	83.95	2.64	-8.72	-9.9							
16	0	83.86	10.25	0.08	-1.47							
17	-2.46	83.69	10.27	0.01	-1.54							
18	0	83.69	10.29	0.07	-1.42							
19	-0.01	83.86	10.21	0.02	-1.48							
20	5.08	83.95	10.24	0.08	-1.48							
21	5.97	83.69	10.21	0.08	-1.56							
22	2.53	83.78	10.32	0.1	-1.43							
23	2.53	83.61	10.26	-0.04	-1.52							
24	0	83.52	10.24	0.08	-1.42							
25	0	83.35	10.31	0.04	-1.46							

Fig.4 Uploaded sensor data from google sheets

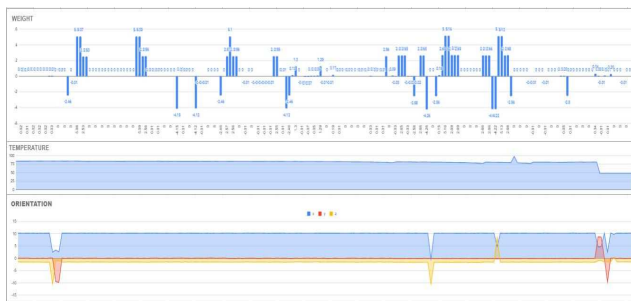


Fig.5 Uploaded sensor values from google sheets

A. Setup

The setup runs and initializes the microcontroller and verifies that it can connect to the cell network, read and tare the load cell, and read data from the MPU 6050. The setup program is shown in fig. 6

```
Starting Arduino web client.
SMS Messages Sender
/tmp/780338060/HX711_GSH_MPU_Relay_Google_Capstone/HX711_GSH_MPU_Relay_Google_Capstone.ino

UNITS: -44103.60
UNITS: 0.07
MPU6050 Setup
Adafruit MPU6050 test!
MPU6050 Found!
SMS to Beekeeper
COMPLETE!
```

Fig. 6 Initialization program of the microcontroller

B. Loop

After setup is complete, it begins to run through the loop and on each iteration uploads the sensor data to the Google sheet. It is shown in fig. 7

```
HX711 Loop
Weight: 0.00
MPU6050 Loop
Temperature 75.98 deg F
Accel X: 10.32 Y: -0.44 Z: -1.11 m/s^2
-----
*
No Tilt
-----
Upload Complete
HX711 Loop
Weight: -0.00 lbs
MPU6050 Loop
Temperature 76.66 deg F
Accel X: 10.29 Y: -0.40 Z: -1.17 m/s^2
-----
*
No Tilt
-----
Upload Complete
```

Fig. 7 Program to upload sensor data to google sheet

C. Conditions and Triggers

If the load cell sensor data is above the upper weight limit it will send a notification to the user that the weight limit has been surpassed and the current weight data. The program is shown in fig. 8

```
MPU6050 Loop
Temperature 78.61 deg F
Accel X: 10.24 Y: -0.34 Z: -1.14 m/s^2
-----
*
No Tilt
-----
Upload Complete
HX711 Loop
Weight: 35.06 lbs
SMS to Beekeeper
COMPLETE!
MPU6050 Loop
Temperature 78.95 deg F
Accel X: 10.22 Y: -0.34 Z: -1.17 m/s^2
-----
*
No Tilt
-----
Upload Complete
HX711 Loop
Weight: -0.01 lbs
```

Fig. 8 Program for upper weight limit

The below weight condition has the same behavior as the upper weight limit but besides the SMS message it also triggers the relay that drives the pump. The below weight limit program is shown in fig. 9

```
HX711 Loop
Weight: -21.69 lbs
SMS to Beekeeper
COMPLETE!
MPU6050 Loop
Temperature 79.37 deg F
Accel X: 10.29 Y: -0.43 Z: -1.22 m/s^2
-----
*
No Tilt
-----
Upload Complete
```

Fig. 9 Program for below weigh limit

The final condition is tilt and it triggers an SMS with the current acceleration values when X acceleration is below 10. The tilt condition program and the text message screenshot are shown in fig. 10 and 11 respectively.

```

MPU6050 Loop
      Temperature 79.63 deg F
      Accel X: 0.68   Y: -0.42   Z: -10.56 m/s^2
-----
*
No Tilt
-----
Upload Complete
Tilt detected Hive needs attention!!
SMS to Beekeeper

COMPLETE!

HX711 Loop
      Weight: 0.01 lbs
MPU6050 Loop
      Temperature 79.63 deg F
      Accel X: 10.22   Y: -0.47   Z: -1.14 m/s^2
-----
*
Tilt detected Hive needs attention!!
-----
Upload Complete
HX711 Loop

```

Fig. 10 Tilt Condition

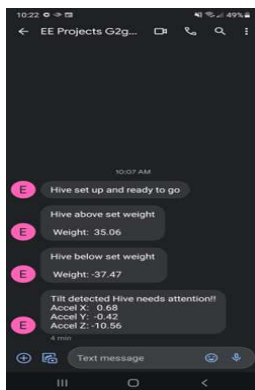


Fig. 11. Tilt message screenshot

V. CONCLUSION

The prototype of the beehive monitor was designed and built successfully. The design was tested with actual beehive the successful data was not collected in googlesheet and SMS. The prototype is capable of demonstrating the viability and advantage of monitoring data from the hives. The design does have some limitations that can be overcome with further development.

There are several improvements that can be made to the prototype. The prototype capabilities can be further improved by adding more sensors. Microphone can be added to pick up hive noise to infer the temperament of the hive. Additional temperature sensors throughout the hive can be added to more accurately detect temperature fluctuations in the hive and detect which part of the hive the bees are congregating in. Humidity sensors can be added inside and outside the hive to bolster temperature data and track worker bee efficiency. The current prototype is limited to only sending out data and could benefit from the ability to receive commands over the internet as well as commands sent over SMS text messaging. Wireless and wired slave devices can be added to collect data from more hives to reduce the cost of scaling the GSM component and costly sim carrier plans. AI and machine learning models could be developed to analyze bee behavior in real time and to predict hive health and behavior.

REFERENCES

- [1] F. Edwards Murphy, E. Popovici, P. Whelan and M. Magno, "Development of an heterogeneous wireless sensor network for instrumentation and analysis of beehives," 2015 IEEE International Instrumentation and Measurement Technology Conference (I2MTC) Proceedings, 2015, pp. 346-351, doi: 10.1109/I2MTC.2015.7151292.
- [2] M. Giammarini, E. Concettoni, C. C. Zazzarini, N. Orlandini, M. Albanesi and C. Cristalli, "BeeHive Lab project - sensorized hive for bee colonies life study," 2015 12th International Workshop on Intelligent Solutions in Embedded Systems (WISES), 2015, pp. 121-126.
- [3] B. Palmer, "Would a World Without Bees Be a World Without Us?," NRDC, 01-Jun-2021. [Online]. Available: <https://www.nrdc.org/onearth/would-world-without-bees-be-world-without-us>. [Accessed: 12-Jun-2021].
- [4] N. Shaghghi, L. Liang, Y. Yabe, S. Lama, J. Mayer and P. Ferguson, "Identifying Beehive Frames Ready For Harvesting," 2019 IEEE Global Humanitarian Technology Conference (GHTC), 2019, pp. 1-4, doi: 10.1109/GHTC46095.2019.9033045.
- [5] M.S. Mahamud, M. A. Al Rakib, T. M. Faruqi, M. Haque, S. A. Rukaia and S. Nazmi, "Mouchak - An IoT Basted Smart Beekeeping System Using MQTT," 2019 4th International Conference on Robotics and Automation Engineering (ICRAE), 2019, pp. 84-88, doi: 10.1109/ICRAE48301.2019.9043815.
- [6] D. W. Fitzgerald, F. E. Murphy, W. M. D. Wright, P. M. Whelan and E. M. Popovici, "Design and development of a smart weighing scale for beehive monitoring," 2015 26th Irish Signals and Systems Conference (ISSC), 2015, pp. 1-6, doi: 10.1109/ISSC.2015.7163763.
- [7] D. Howard, G. Hunter, O. Duran and D. Venetsanos, "Progress towards an Intelligent Beehive: Building an Intelligent Environment to Promote the Well-Being of Honeybees," 2016 12th International Conference on Intelligent Environments (IE), 2016, pp. 262-265, doi: 10.1109/IE.2016.60.
- [8] V. G. Rybin, D. N. Butusov, T. I. Karimov, D. A. Belkin and M. N. Kozak, "Embedded data acquisition system for beehive monitoring," 2017 IEEE II International Conference on Control in Technical Systems (CTS), 2017, pp. 387-390, doi: 10.1109/CTS.2017.8109576.
- [9] N. ANAND, V. B. RAJ, M. S. ULLAS and A. SRIVASTAVA, "Swarm Detection and Beehive Monitoring System using Auditory and Microclimatic Analysis," 2018 3rd International Conference on Circuits, Control, Communication and Computing (I4C), 2018, pp. 1-4, doi: 10.1109/CIMCA.2018.8739710.
- [10] A. Draper, N. Obrusnik, P. Zinemanas, P. Monzón and N. Pérez, "Design and implementation of a remote monitoring system to detect contamination in beehives," 2015 CHILEAN Conference on Electrical, Electronics Engineering, Information and Communication Technologies (CHILECON), 2015, pp. 741-745, doi: 10.1109/Chilecon.2015.7404654.
- [11] A. Shahshahani, X. de Briey, E. Henry and S. Bhadra, "Flexible Printed Capacitive Force Sensors Based Beehive Weight Monitoring System," 2019 IEEE International Flexible Electronics Technology Conference (IFETC), 2019, pp. 1-2, doi: 10.1109/IFETC46817.2019.9073776.
- [12] I. Muller, R. M. de Brito, C. E. Pereira and V. Brusamarello, "Load cells in force sensing analysis -- theory and a novel application," in IEEE Instrumentation & Measurement Magazine, vol. 13, no. 1, pp. 15-19, February 2010, doi: 10.1109/MIM.2010.5399212.
- [13] B. Kim, Y. Noh and D. Jeong, "Implementation of Mobile Based Multi-Function Health Monitoring System," 2013 International Conference on IT Convergence and Security (ICITCS), 2013, pp. 1-4, doi: 10.1109/ICITCS.2013.6717842.
- [14] N. Shine, F. A. Wibisono, E. Anggadajaja and T. Rochadiani, "Food Products Monitoring Machine Using Combinations of Multiple Autonomous Sensors," 2018 10th International Conference on Information Technology and Electrical Engineering (ICITEE), 2018, pp. 152-157, doi: 10.1109/ICITEE.2018.8534860.
- [15] J. Lee, S. Kim, W. Park and J. Bae, "Design of a wearable hand exoskeleton system for evaluation of hand functions," 2017 14th International Conference on Ubiquitous Robots and Ambient Intelligence (URAI), 2017, pp. 578-579, doi: 10.1109/URAI.2017.7992673.
- [16] A. Hassan et al., "Statistical scheme for fault detection using Arduino and MPU 6050," 2019 Prognostics and System Health Management Conference (PHM-Qingdao), 2019, pp. 1-7, doi: 10.1109/PHM-Qingdao46334.2019.8942922.

- [17] A. Karnik, D. Adke and P. Sathe, "Low-Cost Compact Theft-Detection System using MPU-6050 and Blynk IoT Platform," 2020 IEEE Bombay Section Signature Conference (IBSSC), 2020, pp. 113-118, doi: 10.1109/IBSSC51096.2020.9332214.
- [18] A. Sofwan, Sumardi, M. R. Irsyad and Najib, "Measurement Design of Sensor Node for Landslide Disaster Early Warning System," 2018 2nd International Conference on Electrical Engineering and Informatics (ICon EEI), 2018, pp. 86-91, doi: 10.1109/ICon-EEI.2018.8784341.
- [19] H. Mansoorzare, A. Todi, S. Moradian and R. Abdolvand, "A Piezo-Capacitive High-Frequency Resonant Accelerometer," 2020 IEEE International Ultrasonics Symposium (IUS), 2020, pp. 1-4, doi: 10.1109/IUS46767.2020.9251353.
- [20] K. Rao et al., "A High-resolution Area-change-based Capacitive MEMS Accelerometer for Tilt Sensing," 2020 IEEE International Symposium on Inertial Sensors and Systems (INERTIAL), 2020, pp. 1-4, doi: 10.1109/INERTIAL48129.2020.9090016.
- [21] K. Ueno, T. Asai and Y. Amemiya, "Temperature-to-frequency converter consisting of subthreshold mosfet circuits for smart temperature-sensor LSIs," TRANSDUCERS 2009 - 2009 International Solid-State Sensors, Actuators and Microsystems Conference, 2009, pp. 2433-2436, doi: 10.1109/SENSOR.2009.5285436.
- [22] J. Heo and J. Lee, "Temperature Sensor Array for Tactile Sensation Using FBG Sensors," SENSORS, 2006 IEEE, 2006, pp. 1464-1467, doi: 10.1109/ICSENS.2007.355910.
- [23] C. Chiang and F. Chang, "Design of a Calibrated Temperature Difference Sensor Transducer for Monitoring Environmental Temperature Difference Applications," in IEEE Sensors Journal, vol. 16, no. 4, pp. 1038-1043, Feb.15, 2016, doi: 10.1109/JSEN.2015.2495216.
- [24] M. F. Riza and N. S. Salahuddin, "Control Home Devices with Voice Commands via a Smartphone," 2019 Fourth International Conference on Informatics and Computing (ICIC), 2019, pp. 1-7, doi: 10.1109/ICIC47613.2019.8985796.
- [25] H. K. Patel, T. Mody and A. Goyal, "Arduino Based Smart Energy Meter using GSM," 2019 4th International Conference on Internet of Things: Smart Innovation and Usages (IoT-SIU), 2019, pp. 1-6, doi: 10.1109/IoT-SIU.2019.8777490.
- [26] R. K. Singhvi, R. L. Lohar, A. Kumar, R. Sharma, L. D. Sharma and R. K. Saraswat, "IoT Based Smart Waste Management System: India prospective," 2019 4th International Conference on Internet of Things: Smart Innovation and Usages (IoT-SIU), 2019, pp. 1-6, doi: 10.1109/IoT-SIU.2019.8777698.
- [27] A. Taralekar, G. Chouhan, R. Tangade and N. Shardoor, "One touch multi-banking transaction ATM system using biometric and GSM authentication," 2017 International Conference on Big Data, IoT and Data Science (BID), 2017, pp. 60-64, doi: 10.1109/BID.2017.8336574.
- [28] M. M. Islam, S. Surov Tonmoy, S. Quayum, A. R. Sarker, S. Umme Hani and M. A. Mannan, "Smart Poultry Farm Incorporating GSM and IoT," 2019 International Conference on Robotics, Electrical and Signal Processing Techniques (ICREST), 2019, pp. 277-280, doi: 10.1109/ICREST.2019.8644300.
- [29] L. Umar, "Smart level sensor based on thermal resistance measurement with self calibration," 2010 International Conference on Autonomous and Intelligent Systems, AIS 2010, 2010, pp. 1-5, doi: 10.1109/AIS.2010.5547040.
- [30] H. Halvorsen, K. Kaspersen, A. Jonsaas and S. Mylvaganam, "Fusion of Body Sensors' Data and Video Images in Assistive Technology," 2019 IEEE Sensors Applications Symposium (SAS), 2019, pp. 1-6, doi: 10.1109/SAS.2019.8706053.
- [31] S. Abdel-Mageid and R. A. Ramadan, "Efficient deployment algorithms for mobile sensor networks," 2010 International Conference on Autonomous and Intelligent Systems, AIS 2010, 2010, pp. 1-6, doi: 10.1109/AIS.2010.5547017.
- [32] Q. Chen, L. Ai, W. Wei, W. Ma, P. Xie and M. Zhang, "Analysis of Music Representations of Vocal Performance Based on Spectrogram," 2010 6th International Conference on Wireless Communications Networking and Mobile Computing (WiCOM), 2010, pp. 1-4, doi: 10.1109/WICOM.2010.5600580.
- [33] P. Billeschou, C. Albertsen, J. C. Larsen and P. Manoonpong, "A Low-Cost, Compact, Sealed, Three-Axis Force/Torque Sensor for Walking Robots," in IEEE Sensors Journal, vol. 21, no. 7, pp. 8916-8926, 1 April 2021, doi: 10.1109/JSEN.2021.3049947.
- G. Eason, B. Noble, and I. N. Sneddon, "On certain integrals of Lipschitz-Hankel type involving products of Bessel functions," Phil. Trans. Roy. Soc. London, vol. A247, pp. 529-551, April 1955. (*references*)
- [34] J. Clerk Maxwell, A Treatise on Electricity and Magnetism, 3rd ed., vol. 2. Oxford: Clarendon, 1892, pp.68-73.
- [35] I. S. Jacobs and C. P. Bean, "Fine particles, thin films and exchange anisotropy," in Magnetism, vol. III, G. T. Rado and H. Suhl, Eds. New York: Academic, 1963, pp. 271-350.